

CRT Questions - Chapter 7

1. Radius and pi
2. It must have the same name as the class itself.
3. **Private** - Only accessible inside the same class, hidden from other classes
Public - Accessible from anywhere, can be used by other classes
4. The variable radius is most likely declared as private inside the Circle class, because of this the statement `dot.radius = 5;` is invalid because radius is private and cannot be accessed directly from outside the Circle class.

5. a) Roo
b) x
c) getX ()
d) setX (int z)
 calculate ()
e) factor ()
f) Roo ()
g) 1. setX 2. getX 3. Calculate 4. factor 4 IN TOTAL

6. **Class** - is a template that defines variables for the state of an object and methods for an object's behaviour. It does not represent a real thing yet, its just the design.

Object - is a real created thing based on the class, an object stores data and can perform actions and provide communication.

A class is the design. an object is the thing made from that design.

9. a) z
b) y, x
c) y
d) x, z

11. **Method Overloading** - Multiple methods in the same class with the same name but different parameters.

Method Overriding - A subclass provides a new version of a method already defined in the parent class.

Method overloading uses the same method name with different parameters in the same class, while method overriding replaces a parent class method with the same signature in a subclass to change its behaviour.